Lab 1 - Basic R Syntax and Plots

Dylan Renard

Professor Brandon Higgs

Gene Expression and Data Visualization

Johns Hopkins University

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Lab 1:

For this lab, we will be using some basic data manipulation and plotting commands in R. We are working with a data set that is comparing the transcript profiles from peripheral B lymphocytes between patients with systemic lupus erythematosus (SLE) and normal healthy controls. The GEO summary of the data set is as follows:

Systemic lupus erythematosus (SLE) is an autoimmune disease with an important clinical and biological heterogeneity. B lymphocytes appear central to the development of SLE which is characterized by the production of a large variety of autoantibodies and hypergammaglobulinemia. In mice, immature B cells from spontaneous lupus prone animals are able to produce autoantibodies when transferred into immunodeficient mice, strongly suggesting the existence of intrinsic B cell defects during lupus. In order to approach these defects in humans, we compared the peripheral B cell transcriptomes of quiescent lupus patients to normal B cell transcriptomes.

Q1) Data Setup and Loading appropriate libraries

▼ Code

```
# Load core library for visualization
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.3.3
```

Q2) Unzip the text file, and read into R

▼ Code

```
data <- read.table("sle_b_cell.txt",
                   header = TRUE,
```

```
                  row.names = 1)

# Extract the first row
head(data, 1)
```

```
             sle.1    sle.2    sle.3    sle.4    sle.5    sle.6    sle.7
1007_s_at 7.201054 6.946044 6.722417 6.478136 7.657807 7.458431 7.128307
             sle.8    sle.9   sle.10   sle.11   sle.12   sle.13   sle.14
1007_s_at 6.893233 7.417972 6.931443 7.043239 6.477673 7.257562 7.196866
            sle.15   sle.16   sle.17 control.1 control.2 control.3 control.4
1007_s_at 7.215563 6.975056 6.946701  7.001722  7.674048   7.20748  7.237135
          control.5 control.6 control.7 control.8 control.9
1007_s_at  6.819833   7.26124  6.934885  7.003605  7.275157
```

Q3) Look at the dimensions of the data. There should be 26 samples.

▼ Code

```
dim(data)
```

```
[1] 34853    26
```

Q4) Print the sample names to screen.

▼ Code

```
colnames(data)
```

```
 [1] "sle.1"     "sle.2"     "sle.3"     "sle.4"     "sle.5"     "sle.6"
 [7] "sle.7"     "sle.8"     "sle.9"     "sle.10"    "sle.11"    "sle.12"
[13] "sle.13"    "sle.14"    "sle.15"    "sle.16"    "sle.17"    "control.1"
[19] "control.2" "control.3" "control.4" "control.5" "control.6" "control.7"
[25] "control.8" "control.9"
```

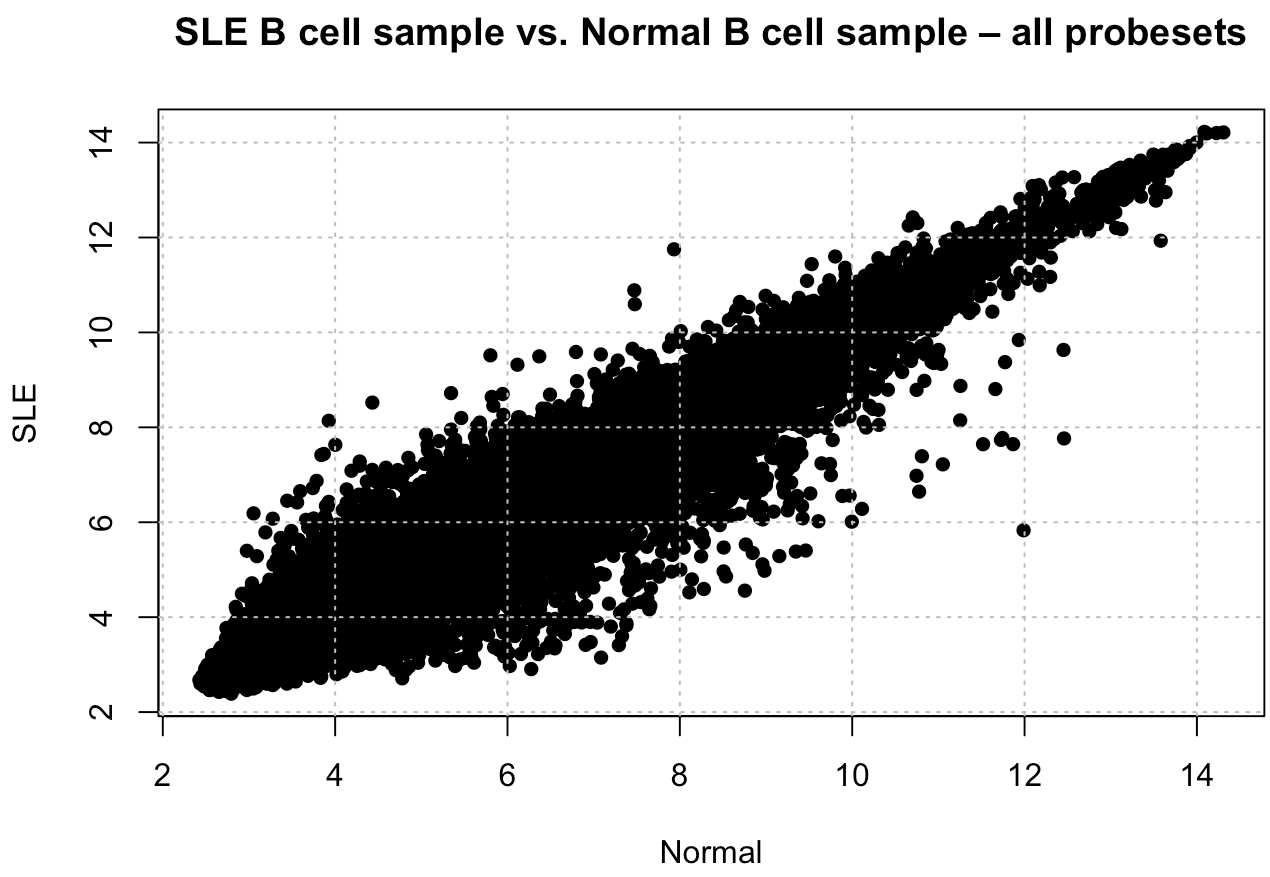
Q5) Basic Scatter Plot across two patients

Plot the second SLE patient sample versus the first normal control samples in an xy scatter plot. Remember that the first argument is the x vector. Label the x and y-axes as 'Normal' and 'SLE', respectively. Title the plot, 'SLE B cell sample vs. Normal B cell sample – all probesets'. Add grey grid lines with the function grid().

▼ Code

```r
# Selecting Subsets of Data
normal1 <- data[, 1]   # first normal control sample
sle2    <- data[, 10]  # second SLE patient sample

# Scatter plot
plot(normal1, sle2,
          xlab = "Normal", ylab = "SLE",
          main = "SLE B cell sample vs. Normal B cell sample – all probesets",
          pch = 16, col = "black")
        # Add grey grid lines
        grid(col = "grey")
```



Q6) Now do the same plot but pick only the first 20 probesets.

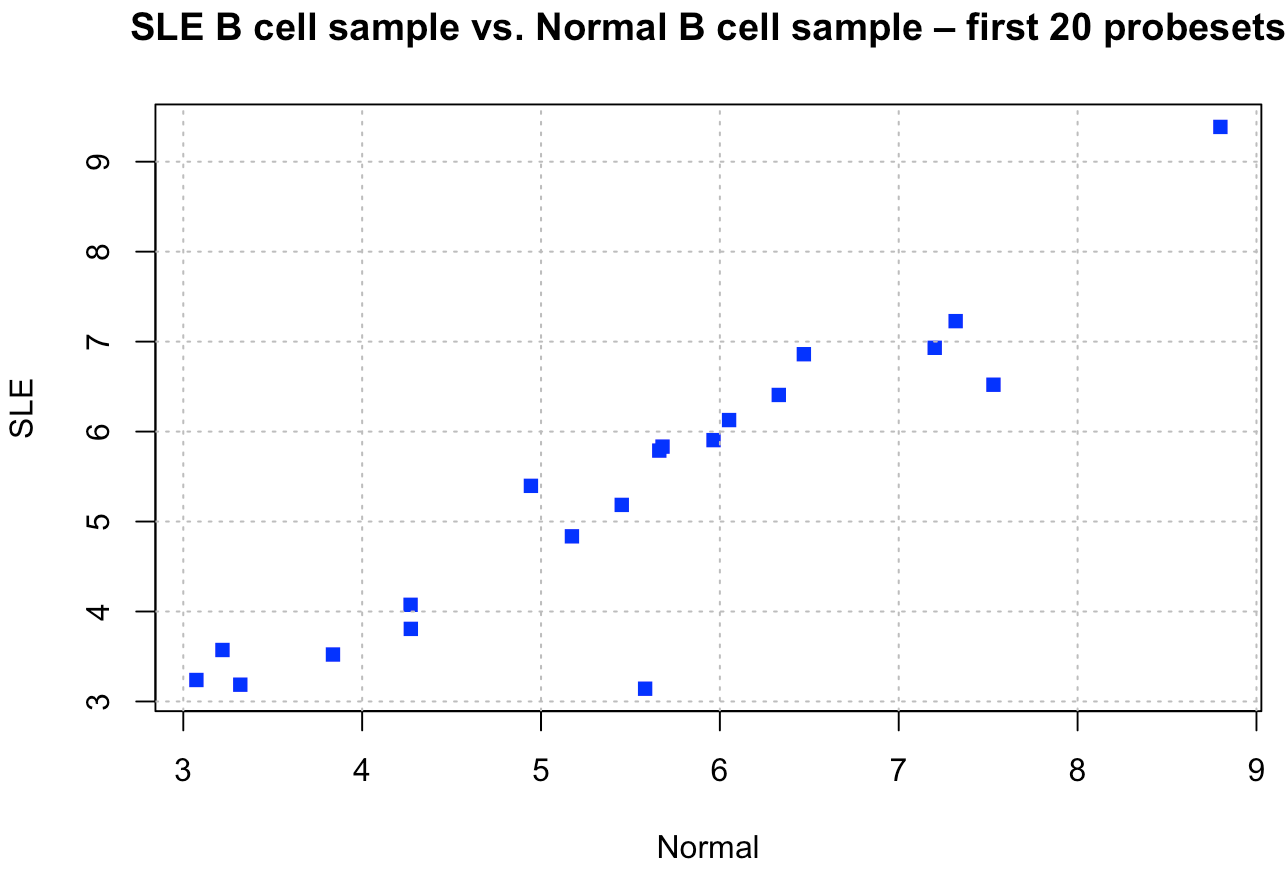
Use the pch=15 argument to change the shape and color the points blue with the col argument.

▼ Code

```r
# First normal control and second SLE sample
normal1 <- data[1:20, 1]   # first 20 rows of Normal sample
sle2    <- data[1:20, 10]  # first 20 rows of SLE sample
```

```
plot(normal1, sle2,
     xlab = "Normal",
     ylab = "SLE",
     main = "SLE B cell sample vs. Normal B cell sample – first 20 probesets",
     pch = 15,           # solid square
     col = "blue")       # color the points blue

grid(col = "grey")
```



Q7) Now plot the following gene in a gene profile plot, IGLJ3 (immunoglobulin lambda joining 3)

which is probeset ID 211881_x_at. This type of plot has the sample indices across the x-axis and the intensities on the y-axis, so you can see a profile of the gene across experiments or arrays. First plot the ranges using the type="n" argument and the plot() function, then add the genes with the lines() function call. Add grid lines.

▼ Code

```
# Extract the expression values for IGLJ3
gene_id <- "211881_x_at"
```

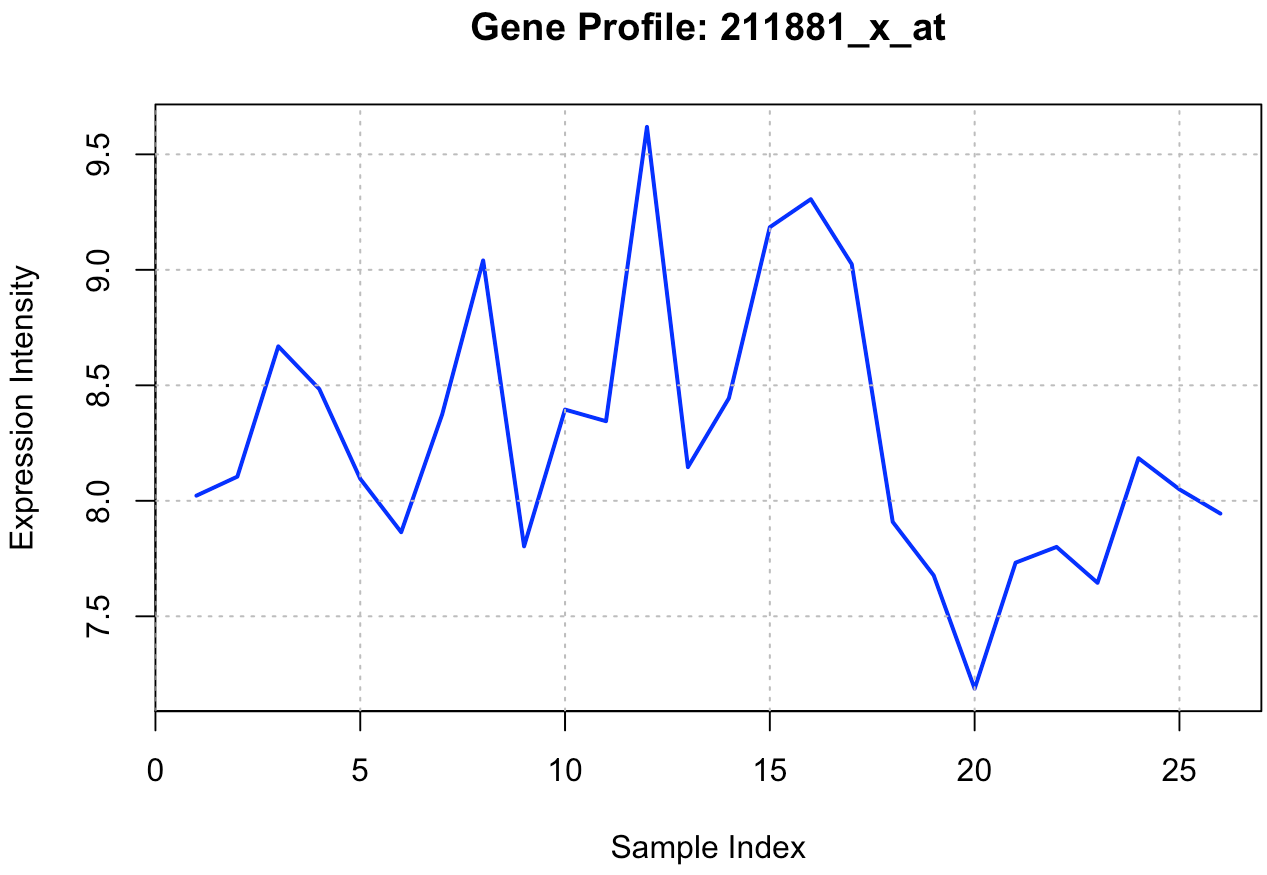
```r
igl_values <- as.numeric(data[gene_id, ])

# X-axis = sample indices (1 through 26)
x_vals <- 1:ncol(data)

# Set up the plot ranges with type = "n" (draws axes but no points/lines yet)
plot(range(x_vals),
     range(igl_values),
     type = "n",
     xlab = "Sample Index",
     ylab = "Expression Intensity",
     main = paste("Gene Profile:", gene_id))

# Add the line plot for IGLJ3
lines(x_vals, igl_values, col = "blue", lwd = 2)

# Add grid lines
grid(col = "grey")
```



Q8) Box Plot question

▼ Code

```r
## Q8 — Boxplot of IGLJ3 (211881_x_at) across SLE vs Control
## Rows = genes (probesets), cols = samples; SLE first (17), then Control (9)

gene_probe <- "211881_x_at"

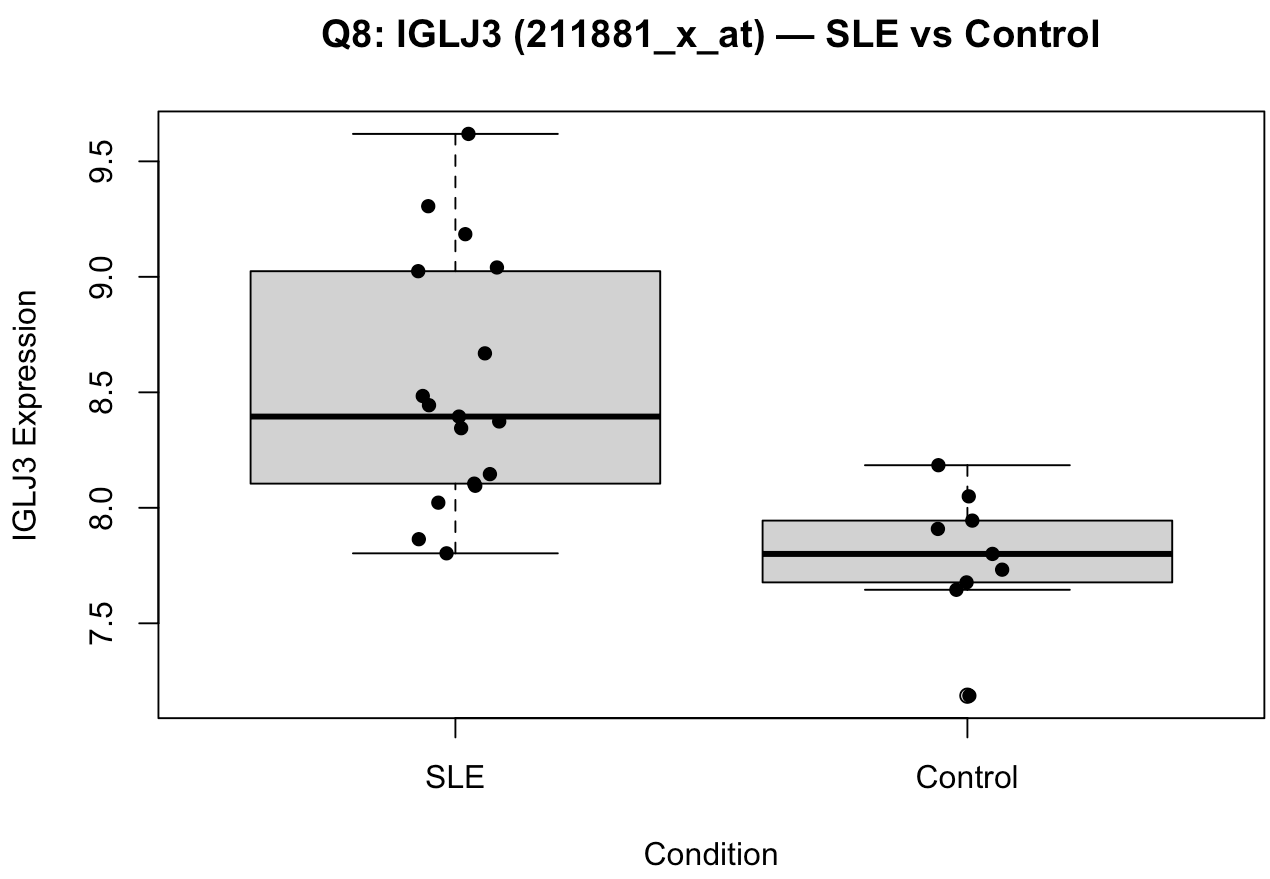
# Expression vector for IGLJ3 across all 26 samples (columns)
y <- as.numeric(data[gene_probe, ])

# Condition factor: 17 SLE first, then 9 Control (and show SLE first on the x-axis)
f <- factor(c(rep("SLE", 17), rep("Control", 9)),
            levels = c("SLE", "Control"))

# Align lengths and drop any NAs
y <- y[seq_len(length(f))]
keep <- !is.na(y)
y <- y[keep]; f <- f[keep]

# Boxplot + optional stripchart overlay
boxplot(y ~ f,
        xlab = "Condition",
        ylab = "IGLJ3 Expression",
        main = "Q8: IGLJ3 (211881_x_at) — SLE vs Control")

with(data.frame(y, f),
     stripchart(y ~ f, method = "jitter", vertical = TRUE, pch = 16, add = TRUE))
```



Sources:

H. Wickham. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York, 2016.

R Core Team (2025). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/.